Joseph Kang

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EDUCATION

University of Pennsylvania (Part Time) August 2026 – May 2028 (Expected)
M.S.E. in Artificial Intelligence, School of Engineering and Applied Science *Philadelphia, PA*

Pennsylvania State University, Schreyer Honors College August 2021 – May 2025
B.S. in Engineering Science & B.S. in Computer Science, summa cum laude *State College, PA*

- Cumulative GPA: 3.99 | Major GPA: 4.00
- Honors Thesis: *Fast and Resource-Efficient Ultrasound Segmentation Using FPGAs*

RESEARCH EXPERIENCE

Undergraduate AI/FPGA Researcher January 2024 – August 2025
Almekkawy Lab, Pennsylvania State University *State College, PA*

- Designed and ran a controlled compression study across three segmentation domains (BraTS brain tumor, Oxford-IIIT Pet, CAMUS cardiac ultrasound), reducing a U-Net from ∼31M to under 60K parameters (>99%) while quantifying accuracy retention (Dice 0.7352–0.8396)
- Synthesized compressed models via hls4ml for the XCU250 FPGA and validated behavior through Vitis Unified Software Platform
- Benchmarked inference latency against CPU, GPU, and TPU baselines (<1 ms and >100× speedup on BraTS and Oxford-IIIT Pet; ∼8 ms and >13× on CAMUS) and isolated DSP block saturation as the binding resource constraint
- Sole first author for resulting paper at IEEE IUS 2025 (Utrecht, Netherlands)

Assistant Electrical Engineer October 2021 – October 2022
Detector Development Lab, Pennsylvania State University *State College, PA*

- Contributed to two NASA-funded astronomical instrumentation missions: BlackCAT (CubeSat wide-field telescope for cosmic transients, led by Prof. Abe Falcone) and tREXS (suborbital sounding rocket for soft X-ray spectroscopy, led by Prof. Randall McEntaffer)
- Chief assembler for the BlackCAT temperature regulator board, a printed circuit board with 300+ components
- Assembled and ran tests on flight PCBs; assisted with rocket integration at NASA Wallops Flight Facility

PROJECTS

Reward-Hacking Forensics on a Reasoning LLM (GitHub) *August 2026 – October 2026*

- Ran a 14-condition prompt-intervention study (3,025 rollouts, full chain of thought) on DeepSeek V4 Flash where an in-context grader rewards defying the user; built the async Python harness and statistics
- Showed gaming tracks how enforceable a check looks, not its stated probability: naming a human reviewer cut it from 92% to 0%, a regex checker only to 36%; stated audit rates of 25–100% were indistinguishable
- Found test-awareness language does not influence compliance: it appears in 76–100% of CoTs in every condition while gaming spans 0–96%

Climbing Data Analysis & Public Technical Communication *March 2026 – Present*

- Scrape and clean public climbing datasets (climbing-history.org, 8a.nu) covering decades of ascent records, handling inconsistent grading scales, sparse athlete histories, and self-reported noise
- Built regression models in scikit-learn to model bouldering grade progression over time and presented results through custom visualizations
- Published analyses as long-form video essays, individually reaching 20–35K viewers (pro-climber ranking, sport climbing progression, bouldering progression)

PUBLICATIONS

- **Joseph Kang**, Ahmed Al-Qurri, and Mohamed Almekkawy, "Fast and Resource-Efficient Ultrasound Segmentation Using FPGAs," in *2025 IEEE International Ultrasonics Symposium (IUS)*, Utrecht, Netherlands, Sep. 2025, doi:10.1109/IUS62464.2025.11201783.
- James H. Tutt, Drew M. Miles, Ross McCurdy, **Joseph Kang**, et al., "Focal plane camera on tREXS," *Journal of Astronomical Telescopes, Instruments, and Systems* 12(2), 026001, 2026, doi:10.1117/1.JATIS.12.2.026001

INDUSTRY EXPERIENCE

Data & AI Program Analyst September 2025 – Present
J.P. Morgan Chase *Wilmington, DE*

- Building a first-of-its-kind unified dataset spanning 125M+ customers and 240M+ accounts across the six lines of business in Consumer Banking, supporting a top-priority initiative from senior leadership
- Built a 100-question evaluation set for an LLM-powered natural language query environment and iterated on domain context, semantic metadata, and question-to-SQL examples, raising accuracy from 35% to 97%
- Delivering cross-LOB insights to dozens of partner teams through SQL and ML-driven analysis, dashboards (Tableau, ThoughtSpot), and custom Python visualizations in Databricks

Software Engineer & Data Scientist, Force For Good April 2026 – Present
J.P. Morgan Chase | Latino Economic Development Center *Washington, DC*

- Selected for a competitive 6–8 month program pairing JPMC engineers with nonprofits to build tech solutions
- Consolidating fragmented client data for a nonprofit serving entrepreneurs and homebuyers through loans and counseling into a single centralized, queryable system
- Designing the data foundation and environment to enable LLM-powered natural language querying, allowing staff to match clients with appropriate loans, counselors, and resources

Data Science Intern Summer 2024 (10 Weeks)
J.P. Morgan Chase *Newark, DE*

- Built an ensemble anomaly detection model (XGBoost + Random Forest) with a live interactive dashboard to flag irregularities in incoming financial reports
- Built an end-to-end automation system to calculate and report monthly business metrics from existing data

Software Engineer & Technology Captain April 2023 – April 2024
Penn State THON *State College, PA*

- Developed think.thon.org, the information management system for event registration, donation tracking, and resource requests behind a record $17M raised in 2023–2024

Software Engineering Intern Summer 2023 (10 Weeks)
Prudential Financial *Newark, NJ*

- Developed TypeScript APIs and microservices for a customer service web platform, enabling Prudential policyholders to query insurance policies, submit changes, and communicate with representatives

ACTIVITIES / LEADERSHIP

Data Science Analyst Committee Member *January 2026 – Present*
J.P. Morgan Chase | Wilmington, DE

- One of two committee members planning community-building events for the firm's Data Science Analyst Program
- Manage a quarterly committee budget and handle expense reporting end-to-end

Piano and Figure Skating Instructor *April 2019 – August 2021*
Self-Employed | West Chester, PA

- Former member of Team USA figure skating and nationally competitive pianist
- Experienced in scheduling, teaching, and leading students of all ages, both in public and private lessons

SCHOLARSHIPS & AWARDS

- President Walker, President Sparks, and Evan Pugh Scholar Awards (top 0.5% in graduating class)
- Dr. Judith A. Todd Copley and Dr. Stephen M. Copley Scholarship in Engineering Science and Mechanics
- Lyle W. and Virginia C. Coffey Scholarship in Engineering Science and Mechanics
- Bronze Medal at the 2023 Penn State Competitive Coding Contest

SKILLS

Core: Python, PyTorch, SQL, Scikit-Learn, Pandas, NumPy, Matplotlib, Databricks, Git, TensorFlow
Other: MATLAB, Verilog/HLS, hls4ml, TypeScript, LLM evals, CoT analysis, Teaching, Fundraising